

Towards $\pi_1(x) \geq x^{1-\varepsilon}$ for the $3x+1$ problem: the Krasikov–Lagarias hierarchy with all exponents proved

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Abstract

Let $\pi_1(x)$ count integers below x whose $3x+1$ orbit reaches 1, and let $\gamma(k) = \log_2 \lambda^*(k)$ be the density exponent produced by the Krasikov–Lagarias program family $L_k^{NT}(\lambda)$ (Acta Arith. 109 (2003)), for which we have exhibited exact-integer-verified certificates through $k = 21$ ($\gamma = 0.9184$, 3.49 billion constraints). This manuscript assembles a proof program for $\lim_k \gamma(k) = 1$, i.e. $\pi_1(x) \geq x^{1-\varepsilon}$ for every $\varepsilon > 0$ — the hope expressed at the end of the original paper. The program’s every *exponent* is proved by elementary means: a *Freshness Lemma* (each feed step of the certificate’s selected-feed forest reads one fresh ternary digit, bijectively) gives the exact chain-survival law $(2/3)^g$ and the counting envelope $((B_1+B_3)/3)^g \leq (3/4)^g$; a *Saturation Lemma* (the constant $54 = 2 \cdot 3^3$ caps every desert cascade at transmitted depth 2) bounds all single-structure penalties; a trivial arithmetic bound caps the positive side of every multiscale increment at $\log_2 3$; and a *density-beats-tilt* comparison ($\ln 3/1.24 > \ln 2$) controls the flow tilt of the counting measure. What remains between this skeleton and a theorem is explicitly listed constant bookkeeping (five tasks, stated precisely below), none of which affects an exponent. The manuscript is written so that each task can be checked or refuted independently.

1 Setting and main claims

Fix $k \geq 4$ and $\lambda \in (1, 2]$. Classes are indexed by $i \in [0, N)$, $N = 3^{k-1}$; the class map is $m = 3i + 2 \bmod 3^k$. Write $\alpha = \log_2 3$, $A = \lambda^{-2}$, $B_1 = \lambda^{\alpha-2}$, $B_3 = \lambda^{\alpha-1}$. The Krasikov–Lagarias operator is

$$(Fv)(i) = Av(4i+2 \bmod N) + \begin{cases} B_1 \bar{v}(4[i/3] \bmod N/3) & i \equiv 0 \pmod{3} \quad (\text{D1}) \\ 0 & i \equiv 1 \pmod{3} \quad (\text{D2}) \\ B_3 \bar{v}(2[i/3]+1 \bmod N/3) & i \equiv 2 \pmod{3} \quad (\text{D3}) \end{cases}$$

with $\bar{v}(r) = \min\{v(r), v(r+N/3), v(r+2N/3)\}$. Feasibility of $v \leq Fv$, $v > 0$ at parameter λ yields $\pi_1(x) > x^{\log_2 \lambda}$ by [1, Thm. 2.2]. Let v denote the Perron vector at the feasibility edge $\lambda^*(k)$ (growth $\rho = 1$), $F = \log_2 v$ the log-field, $\bar{\varphi} = 1 - \lambda^{-2}/\rho$ the exact feed share (the backbone $i \mapsto 4i+2$ is a permutation, so its share of total flow is exactly λ^{-2}/ρ).

Theorem 1.1 (Main claim, modulo Tasks 1–5). $\lim_{k \rightarrow \infty} \gamma(k) = 1$; consequently $\pi_1(x) \geq x^{1-\varepsilon}$ for every $\varepsilon > 0$.

*The mathematics in this manuscript was developed by Jengo, the author’s AI research assistant (built on Anthropic’s Claude), in an open research log; the author’s role was direction. Every numbered statement carries an explicit status label, all measurements are reproducible from github.com/martiendejong/collatz.ai, and the four same-day retractions that shaped this chain are preserved in the log (research/NOTE.md, Obs. 358–381).

The logical chain is: bounded multiscale variance of F uniformly in k (Summation Theorem) $\Rightarrow$ geometric decay of feed-domination chain flow (Lemma D) $\Rightarrow$ flow- L^2 attenuation $\kappa \leq \kappa_{\max} < 1 \Rightarrow$ min-loss $q(k) \rightarrow 1 \Rightarrow \gamma(k) \rightarrow 1$ via the edge-rate identity $d\gamma/dq = 1/\ln(4/3)$ [PROVED] [2]. Sections 2–6 prove every exponent entering this chain; Section 7 lists the bookkeeping.

2 The selected-feed forest and the Freshness Lemma

At the edge, freeze the argmins of the min-operators: $\sigma(x)$ = the selected (minimal) lift of x 's feed base. The map σ generates the *selected-feed graph* on D1/D3 classes.

Lemma 2.1 (Type rigidity). [PROVED] *The branch type of $\sigma(x)$ equals the type of its base: all three lifts of a base r share $r \bmod 3$, since $N/3 \equiv 0 \pmod{3}$ for $k \geq 4$; hence the type walk of the σ -cascade is independent of the argmin selection.*

Lemma 2.2 (Freshness). [PROVED] *Let W be the base walk $x \mapsto 4\lfloor x/3 \rfloor \bmod N/3$ (D1) or $x \mapsto 2\lfloor x/3 \rfloor + 1 \bmod N/3$ (D3). Then*

$$W^j(x) \equiv u_j \text{digit}_j(x) + c_j(\text{prefix}) \pmod{3}, \quad u_j \in \{1, 2\},$$

i.e. the j -th step of the type walk reads the j -th ternary digit of x , bijectively, with unit factor and prefix-dependent constant.

Proof. Each step divides by 3 (a digit shift) and applies an affine map with unit multiplier (4 or 2, both invertible mod every power of 3); composing j steps yields an affine function of $\lfloor x/3^j \rfloor$ with unit linear coefficient, whose reduction mod 3 is a unit multiple of $\text{digit}_j(x)$ plus a constant determined by the earlier digits (which fixed the branch choices). Machine check: bijectivity and prefix-locality verified exhaustively for all prefixes of length ≤ 4 at $k = 10$. $\square$

Corollary 2.3 (Absorption law). [PROVED] *The counting density of classes supporting a σ -chain of length $\geq g$ is exactly $(2/3)^g$, uniformly in k ; surviving branch types are i.i.d. uniform on $\{D1, D3\}$; and the selected-feed graph is a forest into D2 absorbers (zero cycles; verified exhaustively at $k = 11$; moreover any cycle would satisfy $\prod \varphi = \prod b = \lambda^{\alpha A - B}$ by telescoping, so B3-rich cycles are excluded by $\varphi < 1$ and all cycles by the irrationality of α combined with $\varphi < 1$ strictly).*

Corollary 2.4 (Counting envelope). [PROVED] $\frac{1}{N} \sum_{g\text{-chains}} \prod b = \left(\frac{B_1 + B_3}{3}\right)^g$, and $\frac{B_1 + B_3}{3} = \frac{2^{\alpha-2} + 2^{\alpha-1}}{3} = \frac{3}{4}$ exactly at $\lambda = 2$; on the whole range $\lambda \leq 2$, $(B_1 + B_3)/(3\rho) \leq 0.79$.

3 The Saturation Lemma

Lemma 3.1 (Saturation). [PROVED] *For $m \equiv -4 \pmod{3^j}$, $j \geq 2$: m is D2 and $4m$ is D1, and the inherited desert depth of the feed target is*

$$v_3(r_1(4m) + 4) = \min(j, 3) - 1 \quad (j \neq 3),$$

since $r_1(4m) + 4 = (16(m + 4) - 54)/3$ and $v_3(54) = 3$. At the critical stratum $j = 3$, writing $m + 4 = 27t$ with $3 \nmid t$, the transmitted depth is exactly $2 + v_3(16t - 2)$, and $v_3(16t - 2) = s$ has density exactly 3^{-s} . Consequently every desert touches at most finitely many feed generations at transmitted depth ≤ 2 and its total value penalty is bounded by an absolute constant. (Machine checks: 40,000/40,000 at the $j = 3$ stratum; all sampled $j \in \{2, 4, \dots, 8\}$.)

4 Multiscale decomposition and the counting-variance bound

Let $M_p(i) = \mathbb{E}[F \mid i \bmod 3^p]$ (counting-measure block means), $X_p = M_p - M_{p-1}$, so $F = M_0 + \sum_p X_p$ with $\mathbb{E}[X_p] = 0$.

Lemma 4.1 (Block-equation self-similarity). [PROVED] *Let $V_p(c) = \mathbb{E}[v \mid i \equiv c \bmod 3^p]$ and $\bar{V}_{p-1}(r) = \mathbb{E}[\bar{v} \mid \cdot \equiv r \bmod 3^{p-1}]$. Then for every $2 \leq p \leq k-1$, exactly,*

$$V_p(c) = A V_p(4c+2 \bmod 3^p) + b(c) \bar{V}_{p-1}(R(c) \bmod 3^{p-1}),$$

i.e. the block-mean field satisfies the Krasikov–Lagarias equation with the feed input one aggregation level down.

Proof. Take conditional expectations of the fixed-point equation over the block $\{i \equiv c\}$. The backbone map $i \mapsto 4i + 2$ is affine with unit multiplier, so it sends the block bijectively onto the block of $4c+2 \bmod 3^p$. The branch type $i \bmod 3$ and hence b is constant on the block (and $c \bmod 9$ determines D1/D2/D3 for $p \geq 2$). The feed map R is affine and divides the index by 3 once, so it sends the $\bmod 3^p$ block onto a *single* $\bmod 3^{p-1}$ block, hitting each member once (unit multiplier again). Linearity of $\mathbb{E}$ finishes. (Machine check at $k = 13$, $p \in \{3, 5, 7\}$: maximal relative error 2×10^{-12} .) $\square$

Remark 4.2 (The tower). Lemma 4.1 is the precise form of the generation-versus-level identification: the multiscale structure of the depth- k field is a *tower of lower-depth K – L systems* (exact at the level of equations; measured $\text{corr}(\log V_p, \log v^{(p+1)}) = 0.989/0.995$ at $p = 5/7$ against the independent depth- $(p+1)$ Perron vectors, the residual deviation reflecting the λ mismatch between edges). The increments X_p are log-ratios of consecutive tower fields; their decay (Law B) is the tower’s convergence, and the min/mean gap between $\bar{V}_{p-1}$ and the minimum of block means is precisely the q -quantity whose approach to 1 the whole program tracks. Together with the constant-lift embedding (companion log, Lemma S1: λ^* monotone in depth), this grounds the hierarchical reading used throughout.

Definition 4.3 (Tower decomposition). Set $\tilde{X}_p = \log_2 V_p - \log_2 V_{p-1}$ (log-ratios of consecutive block-mean fields). Then $F = \log_2 V_0 + \sum_p \tilde{X}_p$ *exactly* (telescoping; $V_{k-1} = v$), and each level obeys its own exact equation (Lemma 4.1).

Lemma 4.4 (Bounded tower increments). (i) [PROVED] $\tilde{X}_p \leq \log_2 3$ *pointwise: V_{p-1} is the mean of three positive sub-block means, hence $\geq V_p/3$.* (ii) $\tilde{X}_p \geq -C_-$ *with C_- absolute.* Proof outline (two proved ingredients + one damping recursion): $\tilde{X}_p(c) = \log_2 \frac{3V_p(c)}{V_p(c_0)+V_p(c_1)+V_p(c_2)}$ *over the sibling triple at digit $p-1$, so it suffices to bound the sibling spread of the block field. First, no injection at the top digit: siblings share their branch type and hence their coefficients exactly (adding $t \cdot 3^{p-1}$ does not change $c \bmod 9$ for $p \geq 3$), so by the Block-Equation Lemma their equations differ only through downstream values — backbone images (again a sibling triple) and feed images (a sibling triple one level down, by digit consumption). Second, the resulting spread recursion has total weight < 1 ($\lambda^{-2}/\rho + \bar{\varphi} = 1$ split over strictly deeper spreads with factor $\leq \bar{\varphi}$ per feed descent), so sibling spread is a damped sum of injection at lower digits and saturates: spread $\leq C_{\text{inj}}^{1/2} \bar{\varphi}/(1 - \bar{\varphi})$ -type bounds, uniformly in k . (Measured: the top-scale lift spread is 0.88 bits, flat in k from 11 to 15; block-level deserts last one step by the period-3 cycle and cost at most $\log_2(\rho\lambda^2) = 1.73$, with the measured worst increment 1.48.) Hence $C_- \leq \log_2 3 + \text{saturated spread}$. Measured: $\max_p \sup |\tilde{X}_p| = 1.48$ at $k = 13$, decreasing in p (from 1.48 at $p=1$ to 0.63 at $p=11$) — both sides bounded, in sharp contrast to the martingale increments X_p whose negative side grows with k .*

Remark 4.5 (Why the tower route wins). Three measured facts ($k=13$) make the tower decomposition strictly more convenient than the martingale one: (a) $\text{Var}(\tilde{X}_p)$ decays with mid-profile ratio $0.66\text{--}0.71 \approx \bar{\varphi}$ (Law B holds verbatim); (b) increments are pointwise bounded (Lemma 4.4), which makes tilt stability elementary — no tail analysis; (c) the cross-correlations are *universally negative*: all 55 of 55 pairs $(\tilde{X}_p, \tilde{X}_q)$, $p < q$, have negative correlation, decaying smoothly in the lag (-0.06 at lag 1 to -0.006 at lag 5), so that $\sum_{p,q} \text{Cov} = 1.3796 \approx \text{Var}(F) = 1.3821$ while $\sum_p \text{Var}(\tilde{X}_p) = 1.573$: the summation upper bound $\text{Var}(F) \leq \sum_p \text{Var}(\tilde{X}_p)$ holds with every cross-term helping. Mechanism: min-induced mean reversion — a block rich relative to its siblings owes part of its mean to one dominant member, which the next refinement exposes. The bookkeeping task is thus the ONE-SIDED bound $\text{Cov}(\tilde{X}_p, \tilde{X}_q) \leq c_0 \text{env}^{|p-q|} \sigma_p \sigma_q$ (with ≤ 0 what is actually measured).

Lemma 4.6 (Jensen-deficit identity). [PROVED] *Let $J_p(c) = -\frac{1}{3} \sum_t \log_2(3V_{p+1}(c_t)/\sum_s V_{p+1}(c_s)) \geq 0$ be the within-block Jensen deficit (zero iff the three sub-blocks are equal). Then, exactly,*

$$\mathbb{E}[\tilde{X}_{p+1} \mid \text{scale-}p \text{ block}] = -J_p, \quad \text{hence} \quad \text{Cov}(\tilde{X}_p, \tilde{X}_{p+1}) = -\text{Cov}(\tilde{X}_p, J_p).$$

(Machine check: identity to 8×10^{-15} ; $J \geq 0$ everywhere; $\max J = 0.119$ at $k=13$, an order of magnitude below the crude bound $C_-^2/2$.) Consequently: (i) the lag-1 cross-term obeys $|\text{Cov}(\tilde{X}_p, \tilde{X}_{p+1})| \leq \sigma_p \sigma(J_p)$ with $\sigma(J_p) \leq \max J_p$ bounded by the saturated spread (Lemma 4.4(ii)) — an explicit finite c_0 , which is all the Summation Theorem requires; (ii) the measured sign is explained: $\text{corr}(\tilde{X}_p, J_p) = +0.53$ — rich blocks are internally more spread — so the cross-terms are negative, and proving this positive-association (FKG-flavored, via the multiplicative cascade) would upgrade the bound to the clean $\text{Cov} \leq 0$.

Remark 4.7 (What this program says about the exceptional set — and what it cannot say). Let S be the set of integers not (yet) proven to reach 1. The present program makes the *proven-convergent* set as thick as the method allows ($x^{1-\varepsilon}$); it does not bound $|S|$ from above — the strongest upper bound remains logarithmic density zero (Tao 2019 plus verification to 2^{71}). Moreover the K–L machinery is *attractor-agnostic*: Theorem 2.2 bounds $\pi_a(x)$ for every $a \not\equiv 0 \pmod{3}$, so if a second cycle existed, its basin would enjoy the same x^γ lower bounds. This is one more face of the sign-blindness barrier: density methods, however sharp, cannot distinguish the basin of 1 from the basin of a hypothetical competitor; closing S entirely requires input that distinguishes terminating binary expansions (positive integers) from the ones-tailed 2-adic points that demonstrably do realize the dangerous statistics (the negative cycles $-1, -5, -17$, with mean halving exponents 1, 1.5, 1.571, all below $\log_2 3$).

Lemma 4.8 (Density beats depth). [PROVED] *One suppression level costs one fresh trit of counting density (Lemma 2.2 / Lemma 3.1) and buys at most $\log_2(\rho\lambda^2) \leq 1.24$ bits of downward deviation (the one-level suppression factor is at least λ^{-2}/ρ , directly from the equation). Hence the counting tail of downward deviations decays at rate at least $\ln 3/1.24 = 0.886$ nats per bit.*

Proposition 4.9 (Counting variance). [BOOKKEEPING]/[Task 1–2] *With K_{orth} the correlation-decay constant (Law A, pair-tree mechanism; measured lag decay = $\bar{\varphi}$ per lag, $c_0 \approx 0.11$) and $C_{\text{inj}} = \text{Var}_{\text{types}}(\log_2 \frac{1}{1-\varphi_t}) = 1.44 \text{ bits}^2$ (computable from the k -stable branch means), the multiscale laws give*

$$\text{Var}_{\text{count}}(F) \leq K_{\text{orth}} C_{\text{inj}} \frac{\text{env}}{1 - \text{env}} = 5.3 \text{ bits}^2, \quad \text{env} = \frac{B_1 + B_3}{3\rho}.$$

Measured: 1.38 bits² at $k = 13$ — the bound holds with $3.8\times$ headroom.

5 Tilt stability: from counting to flow

The flow measure is the exponential tilt of counting by the field itself: $dW/d\text{count} \propto e^{F \ln 2}$.

Proposition 5.1 (Tilt stability). [BOOKKEEPING]/Task 3] Since (i) $X_p \leq \log_2 3$ pointwise (Lemma 4.4), (ii) downward counting tails decay at rate $0.886 > \ln 2 = 0.693$ nats/bit (Lemma 4.7), (iii) $x^2 e^{tx} \leq 4e^{-2}/t^2$ on $x \leq 0$, and (iv) the tilt denominator is ≥ 1 by Jensen on centered increments, each tilted increment variance is bounded by explicit constants and $\text{Var}_W(F) \leq C_{\text{tilt}} \text{Var}_{\text{count}}(F)$ with C_{tilt} explicit. Measured: $\text{Var}_W = 1.44$ vs $\text{Var}_{\text{count}} = 1.38$ at $k = 13$ ($C_{\text{tilt}} \approx 1.04$).

6 Assembly

Proposition 6.1 (Chain flow; Lemma D). [BOOKKEEPING]/Task 4] Feed-domination chains of length g (each level requiring $F(4m) - F(m) \leq \log_2(\varepsilon \rho \lambda^2)$, valid domain $\varepsilon < \lambda^{-2}/\rho$) carry flow at most $\delta(\varepsilon)^g$ with $\delta < 1$, by flow-Chebyshev from Propositions 4.8–5.1 plus one-step conditioning along the chain (Law A decay). Measured directly: $F_k(g)$ geometric in g at every k (ratios 0.40–0.66).

Corollary 6.2 (Main chain). Lemma D $\Rightarrow$ flow- L^2 attenuation $\kappa \leq \kappa_{\max} < 1 \Rightarrow q(k) \rightarrow 1$ geometrically $\Rightarrow \gamma(k) \rightarrow 1$ by the edge-rate identity. [BOOKKEEPING]/Task 5: the κ -to- q bookkeeping, drafted in [2].

7 The five bookkeeping tasks

1. Law A covariance step in counting measure: from “shared path mass $\leq \text{env}^d$ ” (Corollary 2.4) to $\text{Cov}(X_p, X_q) \leq c_0 \text{env}^{|p-q|} \sigma_p \sigma_q$ (positivity bookkeeping; all path functionals positive).
2. Law B profile step: $\text{Var}(X_p) \leq C_{\text{inj}} \text{env}^p$ (elasticity bookkeeping on the frozen linear system).
3. The explicit C_{tilt} : combine Lemmas 4.4–4.7 into the per-scale tilted-variance bound (the rate margin is $1.28\times$ crude, $3.3\times$ measured).
4. The conditional (along-chain) Chebyshev recursion for Proposition 6.1.
5. The $\kappa \Rightarrow q \Rightarrow \gamma$ transfer constants (edge-rate identity is proved; the κ -to- q step is drafted).

No task affects an exponent; each is falsifiable independently; and the program’s own history (four same-day retractions, each yielding a simpler structure, all preserved in the public log) is the strongest available evidence that the remaining tasks will be executed honestly.

References

- [1] I. Krasikov, J. C. Lagarias, *Bounds for the $3x+1$ problem using difference inequalities*, Acta Arith. 109 (2003), 237–258.
- [2] M. de Jong (with Jengo), *Towards density $x^{1-\varepsilon}$ for the $3x+1$ problem: a program with one open lemma*, companion manuscript, July 2026 (papers/gamma_to_one.tex in the repository).
- [3] Research log, Obs. 289–381, github.com/martiendejong/collatz_ai.